MILITARY SPECIFICATION

CALCIUM CHROMATE, TECHNICAL

1. SCOPE

1.1 This specification covers technical grade calcium chromate for use in pyrotechnic ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard for Testing Purposes

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1169- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1233- Procedures for Determining Particle Size, Particle Size Distribution and Packed Density of Powdered Materials
MIL-STD-1234- Pyrotechnics: Sampling, Inspection and Testing
MIL-STD-1235- Single and Multilevel Continuous Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-P77 - Atomic Absorption Spectrometry, Rec. Practices for, Proposed
ASTM-B212 - Apparent Density of Metal Powders, Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Penn. 19103).

3. REQUIREMENTS

3.1 The material shall conform to the chemical requirements shown in Table I.

TABLE I - CHEMICAL REQUIREMENTS

TEST	REQUIREMENT	TEST PARAGRAPH
Calcium chromate, percent, minimum	96.0	4.4.2
Calcium carbonate, percent, maximum	1.5	4.4.3
Chloride (as Cl-), percent, max.	0.01	4.4.4
Sulfate (as SO_4), percent, max.	0.01	4.4.5
Volatile matter, percent, max.	0.10	4.4.6
Aluminum (as Al_2O_3), percent max.	0.20	4.4.7
Sodium (as Na_2O), percent, max.	0.15	4.4.8
Silicon (as Si), percent, max.	0.05	4.4.9

3.2 Apparent density.-The technical calcium chromate shall have an apparent density of 0.35 gm/ml, min. when tested as specified in 4.4.10.

3.3 Particle size.-The particle size of the technical calcium chromate shall be 2.0 $\pm$ 0.5 microns when tested as specified in 4.4.11.

3.4 Granulation.-The technical calcium chromate shall be able to pass, 100 percent, through a 44 micron (No. 325) sieve when tested as specified in 4.4.12.

3.5 First article testing.-This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 50 lbs. of calcium chromate in accordance

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with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of one or more batches of calcium chromate produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of calcium chromate that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

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4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect, except where 100 percent inspection is specified.

4.3.2.1 Containers, prior to sealing

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Liner pierced or torn	Visual	01001
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or illegible	Visual	01002

4.3.3 Sampling.-All sampling of individual containers shall be accomplished in such a manner that the sample is representative of the entire contents of the container. Individual or composite samples shall be identified in such a manner as to provide positive identification as to the container or containers represented by each sample. For the chemical composition tests, the lot shall be divided up into two (2) parts. From each part, a composite sample shall be prepared in the following manner. A random sample of filled containers shall be selected in accordance with MIL-STD-105. However, if each part comprises less than five (5) containers, all containers shall be selected. Approximately equal portions shall be taken from the selected containers in such quantities that the aggregate weight of such portions shall be approximately five (5) pounds (lbs.). These portions shall then be thoroughly mixed to form a composite sample. For the physical tests, the lot shall be divided up into four (4) parts. For each part, a composite sample shall be prepared in the same manner as described above. The aggregate weight of each composite sample shall be approximately ten (10) lbs. If any of the above samples fails to meet any of the requirements specified in Section 3, the lot shall be rejected.

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4.4 Test methods and procedures

4.4.1 Reagents.-The reagents used in testing should conform to the American Chemical Society (ACS) Specifications as described in "Reagent Chemicals".

4.4.2 Determination of calcium chromate - Major defect - Code No. 02001.-Weigh accurately, approximately 2.5 gm of sample. Transfer to a 250 ml volumetric flask, add 20 ml of concentrated hydrochloric acid, and shake to dissolve. The solution may not be entirely clear. Dilute with water to the mark and mix well. Using a 25 ml. volumetric pipette, transfer an aliquot to an iodine flask. Dilute with 100 ml. of water, add 3 gms. of potassium iodide and allow to stand in the dark for 10 minutes. Titrate the liberated iodine with N/10 sodium thiosulfate using starch indicator toward the end. Calculate the percent of calcium chromate as follows:

$$\frac{\text{Ml of Sodium Thiosulfate X Normality X 0.05203 X 100}}{\text{Weight of Sample in Aliquot}} = \%\text{Calcium Chromate "A"}$$

4.4.3 Determination of calcium carbonate - Major defect - Code No. 03001.

4.4.3.1 Reagents

4.4.3.1.1 Ammonium oxalate saturated solution.-Dissolve 50 gms. of ammonium oxalate in 500 ml of hot water, dilute to 1 liter and allow to cool to room temperature.

4.4.3.2 Procedure.-Using a 25 ml. volumetric pipette, transfer an aliquot of the solution from the preceeding test to a 400 ml. beaker. Dilute with water to a volume of about 75 ml. Heat to boiling and add slowly 100 ml. of a saturated solution of ammonium oxalate. Add 2 to 3 drops of methyl red indicator solution. Then add ammonium hydroxide dropwise until the solution just turns yellow, and set on a steam bath until precipitate settles. Filter the precipitate through a Gooch or sintered glass filter and wash well with cold water (100 - 150 ml.). Transfer the filter plus precipitate to the precipitating beaker and cover with water. Stir the filter thoroughly with a glass rod to loosen the precipitate. Add 3 ml. of concentrated sulfuric acid and heat to 70 - 80°C. Titrate with N/10 potassium permanganate to the first pink endpoint, making sure that the temperature is still at 70 - 80°C when the endpoint is reached. Calculate the percent of calcium carbonate as follows:

$$\frac{\text{Ml of Potassium Permanganate X Normality X 0.02004 X 100}}{\text{Weight of Sample in Aliquot}} = \%\text{Calcium "B"}$$

"A" (from 4.4.2) X .2568 = % Calcium, from chromate side = "C"

If "B" is greater than "C", the sample contains free $CaCO_3$ then:
"B" - "C" = % Calcium due to carbonate = "D"
"D" X 2.5 = $\%CaCO_3$

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4.4.4 Determination of chloride-Major defect - Code No. 04001.-The chloride determination shall be performed in accordance with Method 401.1 of MIL-STD-1234. The leeching solvent shall be 20% (by volume) sulfuric acid.

4.4.5 Determination of sulfate - Major defect - Code No. 05001.-The sulfate determination shall be performed in accordance with Method 411.1 of MIL-STD-650. The leeching solvent shall be 20% (by volume) hydrochloric acid.

4.4.6 Volatile matter - Major defect - Code No. 06001.-Heat a glass-stoppered moisture dish in an oven at 100 to 110°C. for 1 hour, cool in a desiccator, and weigh to the nearest 0.1 milligram (mg.). Transfer approximately 2 grams (gm.) of the specimen to the dry dish, stopper and weigh. Heat unstoppered in an oven at 100° to 110°C for 1 hour, cool in a desiccator, replace the stopper, and weigh to the nearest 0.1 mg. Repeat the heating for half-hour periods until successive weighings differ by not more than 0.2 mg. Calculate the loss in weight as percent volatile matter as follows:

$$\text{Percent volatile matter} = 100 \times \frac{A - B}{A - D}$$

Where:

A = weight of original specimen and dish, in gm.
B = weight of specimen and dish after drying to constant weight in gm.
D = weight of dry moisture dish, in gm.

4.4.7 Determination of aluminum (as Al_2O_3) - Major defect - Code No. 07001.-Aluminum shall be determined by an atomic absorption method as specified in ASTM-P77.

4.4.8 Determination of sodium (as Na_2O) - Major Defect - Code No. 08001.-Sodium shall be determined by an atomic absorption method as specified in ASTM-P77.

4.4.9 Determination of silicon (as Si) - Major defect - Code No. 09001.-Silicon shall be determined by an atomic absorption method as specified in ASTM-P77.

4.4.10 Apparent density

4.4.10.1 Primary method - Major defect - Code No. 10001.-The apparent density of technical calcium chromate shall be determined as specified in ASTM-B212.

4.4.10.2 Alternate method - Major defect - Code No. 10002.-The apparent density of technical calcium chromate shall be determined as specified in Method 201.1 of MIL-STD-650.

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4.4.11 Particle size - Major defect - Code No. 11001.-The particle size shall be determined in accordance with Method 100 of MIL-STD-1233.

4.4.12 Granulation - Major defect - Code No. 12001.-Dry a watchglass at 100 to 105°C for 20 minutes, cool in a desiccator and weigh. Weigh approximately 100 gm of the specimen and transfer to the specified sieve, conforming to Specification RR-S-366, which has been placed on an evaporating dish of about 9 inches in diameter. Wash the material through the sieve with water containing a small amount of a wetting agent. A camel's hair brush may be used to facilitate washing the particles through the screen. When no more material will pass through the sieve, wash the residue with acetone. Dry the residue in air, transfer it carefully to the weighed watchglass, dry at 100 to 105°C for 20 minutes, cool in a desiccator, and weight. Calculate the percent of material passed through the sieve as follows:

$$\text{Granulation (percent through)} = 100 \left(1 - \frac{A}{W}\right)$$

Where:
A = weight of residue on sieve, in gm.
W = weight of specimen, in. gm.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A - Not Applicable

5.1.2 Level B - Not Applicable

5.1.3 Level C - Calcium chromate shall be packaged and packed in accordance with the manufacturer's normal practice and in a manner which will insure undamaged delivery of the material to the receiving activity.

5.2 Marking.-Marking shall be in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this document
b. Data cards shall be prepared for each lot in accordance with Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
ARMY-MU

PROJECT NUMBER: 6810-A-921

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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